Metadata: StormSense Comma Separated Values (CSV) for 72-Hour Forecasts (fore72)

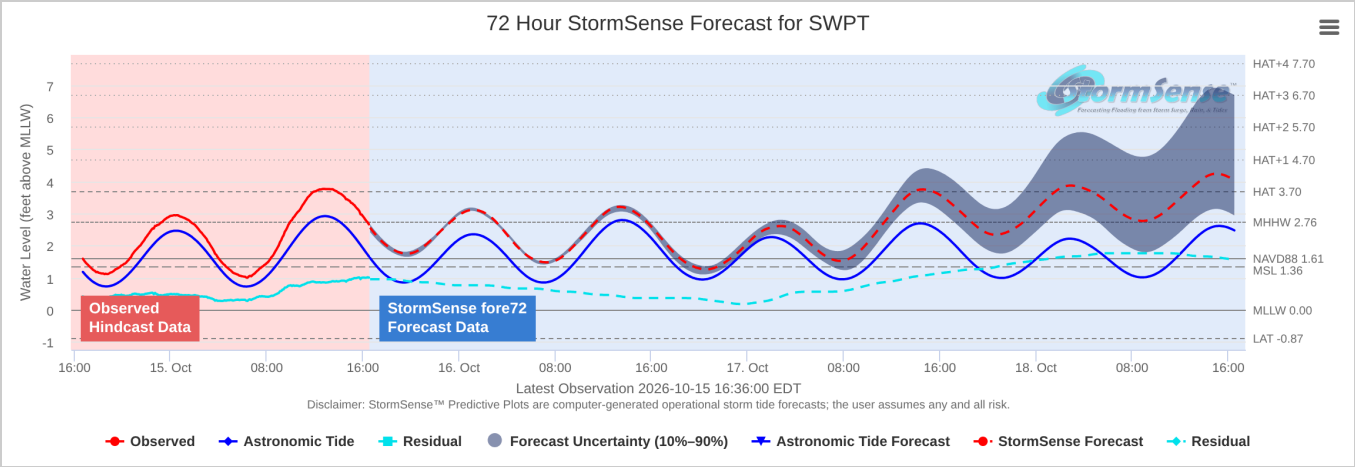


Figure 1. *Example StormSense Chart of 72-hour forecast data (MLLW) at NOAA's Sewells Point tide gauge, in Norfolk, VA, issued October 15, 2026 at 16:36 EDT (2026-10-15T20:36:00Z) from the P-ETSS 12Z cycle.*

Fields **fore72 data downloads only include data series for the 72-hour forecast period (dashed lines and shaded cone)**

date_time — Date and time in Zulu/GMT, formatted as a string leading with the date as "YYYY-MM-DD" followed by the time (T) as "hh:mm:ss" relative to Zulu (Z) time. Recorded in regular 6-minute intervals (0, 6, 12 … 54 minutes past the hour), listed in chronological order.

astro_tide — (**Astronomic Tide Forecast**) Harmonically predicted astronomical tide for the next 72 hours from the NWS P-ETSS model. Formatted in units of feet rounded to three decimal places (thousandth of a foot) relative to Mean Lower Low Water (MLLW).

surge — Ensemble-mean storm surge from P-ETSS (the Probabilistic Extra-Tropical Storm Surge model of the NWS Meteorological Development Laboratory, run with an ensemble of weather forecasts): the water level change driven by forecast wind and air pressure. Hourly guidance interpolated to 6 minutes and smoothed with a 3-hour low-pass filter. Feet; a height difference, so the same in MLLW and NAVD88.

bias — The observation-based correction P-ETSS adds to its surge, from the recent difference between the gauge's observations and the modeled tide plus surge; it is held or decays through the forecast, and is 0.000 where P-ETSS applies no correction. Feet; the same in MLLW and NAVD88.

residual — (**Residual**) Forecast residual: the non-tidal part of the forecast, SSforecast − astro_tide (surge + bias). Feet; the same in MLLW and NAVD88.

SSforecast — (**StormSense Forecast**) The StormSense 72-hour storm tide forecast, the ensemble-mean total water level: astro_tide + surge + bias. Formatted in units of feet rounded to three decimal places (thousandth of a foot) relative to Mean Lower Low Water (MLLW).

SSforecast_p10 — (**Forecast Uncertainty (10%–90%)**) 10th percentile of the ensemble total water level (P-ETSS's 90% exceedance level): the lower edge of the shaded cone of uncertainty. Formatted in units of feet rounded to three decimal places (thousandth of a foot) relative to Mean Lower Low Water (MLLW).

SSforecast_p90 Upper edge of the cone: the 90th percentile of the ensemble total water level (P-ETSS's 10% exceedance level). Nine in ten ensemble members fall at or below it. Formatted in units of feet rounded to three decimal places (thousandth of a foot) relative to Mean Lower Low Water (MLLW).

surge_p10, surge_p90 10th and 90th percentiles of the ensemble storm surge (P-ETSS 90% and 10% exceedance). Feet; the same in MLLW and NAVD88.

P-ETSS levels are tied to each gauge's own record: the median difference between the observations and P-ETSS over the 24 hours before the forecast starts is added to astro_tide, SSforecast and its percentiles (0.00 ft at most NOAA gauges). At stations without a P-ETSS point, the file keeps the same columns with the 36-hour forecast in astro_tide, residual and SSforecast and the P-ETSS-only columns left blank.

Table 1. *Example StormSense Table of 72-hour forecast data (MLLW) at NOAA's Sewells Point tide gauge, in Norfolk, VA, issued October 15, 2026 at 16:36 EDT (2026-10-15T20:36:00Z).*

date_time	astro_tide	surge	bias	residual	SSforecast	SSforecast_p10	SSforecast_p90	surge_p10	surge_p90
2026-10-15T20:36:00Z	1.610	1.000	-0.052	0.948	2.557	2.417	2.657	0.860	1.100
2026-10-15T20:42:00Z	1.560	1.000	-0.052	0.948	2.507	2.372	2.607	0.864	1.100
2026-10-15T20:48:00Z	1.512	1.000	-0.052	0.948	2.459	2.328	2.559	0.869	1.100
2026-10-15T20:54:00Z	1.465	1.000	-0.052	0.948	2.413	2.286	2.512	0.873	1.100
2026-10-15T21:00:00Z	1.418	1.000	-0.052	0.948	2.367	2.244	2.466	0.877	1.099
2026-10-15T21:06:00Z	1.374	1.000	-0.052	0.948	2.322	2.203	2.421	0.882	1.099
2026-10-15T21:12:00Z	1.331	1.000	-0.051	0.949	2.279	2.164	2.377	0.885	1.098
2026-10-15T21:18:00Z	1.289	1.000	-0.051	0.949	2.237	2.126	2.334	0.889	1.097
2026-10-15T21:24:00Z	1.249	1.000	-0.051	0.949	2.197	2.089	2.293	0.891	1.096
2026-10-15T21:30:00Z	1.210	1.000	-0.051	0.949	2.158	2.052	2.252	0.894	1.094
2026-10-15T21:36:00Z	1.174	1.000	-0.051	0.949	2.123	2.018	2.214	0.896	1.091
2026-10-15T21:42:00Z	1.138	1.000	-0.051	0.949	2.088	1.985	2.176	0.897	1.089

Datums *(gray lines labeled at the right of the chart; click a label to hide or show its line)*

····· HAT+1 … HAT+4 Reference lines 1, 2, 3 and 4 feet above HAT, for reading how far an extreme water level rises above the highest predicted tide. Where a station has no published HAT, the steps start from MHHW.

– – – HAT Highest Astronomical Tide: the highest water level predicted from the tide alone (no weather) over the National Tidal Datum Epoch (NTDE, 1983–2001).

- - - - MHHW Mean Higher High Water: the average of the higher of each day's two high tides over the NTDE; a common threshold for nuisance (high-tide) flooding.

—— NAVD88 North American Vertical Datum of 1988: the fixed, land-based geodetic datum used for surveyed elevations, roads and flood maps.

– – – MSL Mean Sea Level: the average of hourly water levels over the NTDE.

—— MLLW Mean Lower Low Water: the average of the lower of each day's two low tides over the NTDE. It is zero on these charts and in these CSVs.

– – – LAT Lowest Astronomical Tide: the lowest water level predicted from the tide alone over the NTDE.

Datum source: at NOAA-operated tide gauges the datums are NOAA's published tidal datums for that station. USGS, VDEM and StormSense sensors natively report live water levels in NAVD88 only, so their MLLW forecasts and observations are datum-shifted from NAVD88 using the MLLW–NAVD88 separation derived from NOAA VDatum

(https://vdatum.noaa.gov/vdatumweb/), and their tidal datum lines come from VDatum as well. Not every datum is available at every station.